A Data-Free PINN Framework for Static Modeling of Magnetically Steered Catheters

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***Abstract*—This research presents a data-free Physics-Informed Neural Network (PINN) to model magnetically steered catheter deformation. The framework embeds the Cosserat rod theory directly into its loss function, enforcing physical constraints without any training data. The model is trained solely on the governing physics equations and boundary conditions. It achieves sub-millimeter accuracy, with evaluations showing errors as low as 0.0182 mm and inference was approximately 450 times faster than the numerical boundary-value problem (BVP) solver for the reference case. This demonstrates strong potential for radiation-minimized surgical navigation.**

***Index Terms*—Physics-informed neural networks, Cosserat rod model, magnetically steered catheters, minimally invasive surgery, boundary value problems, continuum robotics.**

I. Introduction

Minimally invasive surgery (MIS) has revolutionized modern surgery by improving patient outcomes and shortening recovery times [1]. Compared to open procedures, MIS reduces blood loss, soft-tissue trauma, and postoperative recovery time [2]. For instance, one review highlights MIS's minimal invasiveness, improved safety, and expedited recovery [3], leading to better outcomes across specialties [4]. Globally, MIS adoption has increased; one series reports cases rising from 20.4% in 2015 to 57.3% by 2022 [2]. Technological advances, particularly robotic-assisted systems, enhance dexterity and three-dimensional vision, expanding MIS applications [4]. Over the past decade, robotic-assisted surgery has enabled MIS in various fields, improving precision and ergonomics [5]. Surgical robotics therefore promotes MIS and offers clear benefits over open approaches [3].

MIS relies on specialized instruments operating through small incisions, including capsule endoscopes and catheter systems. Endoscopic capsules represent a pivotal technology in minimally invasive procedures, enabling traversal of complex anatomical structures by conforming to the body's contours [6], [7]. Modern capsules include cameras, illumination, and power sources, enabling visualization of otherwise inaccessible intestinal segments with minimal discomfort and risk [8]. Similarly, catheters and guidewires are essential for vascular interventions. In procedures such as percutaneous coronary or neurovascular interventions, a flexible guidewire or catheter is introduced through an access puncture and navigated under X-ray guidance, with manual steering via fluoroscopy [9], [10]. These devices enable therapies such as angioplasty, stenting, and ablation without open surgery. Magnetically actuated catheter robots embed magnets for remote steering using external fields, promising improved maneuverability along complex paths without direct contact [11].

Precise control of catheter shape and tip position remains a critical requirement. Conventional methods often rely on continuous fluoroscopy, exposing patients and clinical staff to ionizing radiation [12]. Over 80% of many MIS interventions depend on fluoroscopic imaging, which entails known hazards [11]. Hence, reducing fluoroscopy usage is highly desirable. Accurate modeling can predict catheter shape from applied forces or control inputs, reducing the need for continuous imaging [13]. For instance, one study on magnetically actuated robots reports reduced procedure time and radiation exposure by minimizing manual adjustments [14]. Thus, predicting tip location from inputs supports navigation with less frequent X-ray feedback.

Catheter modeling commonly assumes quasi-static conditions: motions are slow and inertial effects are negligible. Insertions typically involve low accelerations, so the catheter shapes satisfy static equilibrium equations [15]. This assumption is validated in prior work; several studies use quasi-static torque–deflection segment models that match experiments [9], [16]. Static Cosserat rod formulations capture bending and twisting behavior and simplify simulation to ordinary differential equations (ODEs) along the catheter length.

Machine learning, particularly Physics-Informed Neural Networks (PINNs), offers new avenues for solving physical models. PINNs approximate solutions of ODEs and PDEs by embedding residuals into the loss function [17]. A PINN outputs field variables as functions of coordinates and penalizes violations of governing equations and boundary conditions, potentially without large labeled datasets. PINNs are mesh-free, can integrate data when available, and often provide favorable scaling between physics enforcement and data efficiency [18].

This study employs a PINN framework to solve the static Cosserat equations for magnetic catheters. The model tracks centerline position and orientation by enforcing force and moment balance with magnetic torques. The effectiveness of the approach is evaluated in both two- and three-dimensional

configurations through a comparative study between a conventional numerical boundary value problem (BVP) solver and a PINN.

In summary, this research addresses a key challenge in minimally invasive surgery: reducing reliance on fluoroscopy for navigating magnetically steered catheters. Accurate prediction of the catheter's shape is essential. We use a PINN to solve the physical equations governing the catheter's static deformation under magnetic forces, and the performance of this approach is compared against a traditional numerical BVP solver for both 2D and 3D scenarios.

II. Previous Work

Physics-informed neural networks (PINNs) have gained significant traction in modeling continuum robotic systems and soft structures. Bensch et al. implemented a PINN for the static Cosserat rod formulation of a tendon-driven robot [19], embedding the governing equations into the loss function together with boundary information derived from a shooting-method reference. Their model reproduced robot configurations with high efficiency, achieving a median positional error of approximately 1 mm ($\approx$ 1% of the total length) for a 20-segment system. Licher et al. proposed a domain-decoupled PINN surrogate for Cosserat dynamics in model predictive control [20]. Trained using the governing equations and Kalman-filtered tip measurements, their framework predicted strain and velocity profiles with tracking errors of 2–3 mm ($\approx$ 2–3% of the length). These studies demonstrated the capability of PINNs to capture the mechanics of Cosserat rods; however, their accuracy remained dependent on boundary data or sensor-based supervision.

Most existing PINN frameworks for continuum robots are therefore partially data-driven, combining physics residuals with supervised loss components. In particular, Bensch et al. relied on boundary conditions generated by a numerical solver [19], while Licher et al. incorporated feedback from measurement data [20]. Consequently, their formulations included explicit data discrepancy terms in addition to PDE and boundary residuals. In contrast, the present study introduces a fully physics-driven PINN that eliminates all supervised components. The network is trained exclusively on the governing equations and wrench-type boundary conditions, without requiring sample points, sensor data, or precomputed trajectories—thereby exploiting a principal advantage of PINNs: minimal dependence on data and mesh discretization.

While previous data-dependent approaches achieved millimeter-scale accuracy—approximately 1 mm in [19] and 3 mm in [20]—the proposed data-free framework attains sub-millimeter tip errors against a numerical BVP solution obtained for the same governing equations and parameter configuration, corresponding to only a few hundredths of a percent of the robot's total length, without any measurement supervision. To the best of our knowledge, this constitutes the first demonstration of sub-millimeter accuracy in catheter- or continuum-rod modeling achieved solely through physics-based learning.

In summary, prior PINN studies advanced Cosserat-based modeling and control by integrating physical and data-driven terms but remained constrained by their reliance on external data. The proposed approach removes all supervised loss components and achieves superior precision purely through physics-consistent training. Comparison with numerical boundary value problem (BVP) solutions further confirms the accuracy and robustness of the presented fully physics-informed, data-free PINN framework for continuum mechanics applications.

III. Cosserat Equations & System Specifications

In practical applications, catheters are manipulated with slow movements, rendering quasi-static modeling appropriate. Consequently, the governing equations are based on the Cosserat rod model, with all temporal dependencies eliminated to reflect the quasi-static nature of the system. The resultant equations are solely partial derivatives with respect to the arc-length parameter s, accurately capturing large deformations, spatially varying curvature, and a geometrically exact representation of the catheter's shape under magnetic actuation.

Till et al. employed the Cosserat rod framework to simulate the dynamics of continuum robots. They discretized the time derivatives implicitly, solving a boundary value problem in arc length at each timestep. The authors note that under quasi-static assumptions, the dynamic equations reduce to static equilibrium equations. Therefore, the quasi-static Cosserat equations are as follows [21]:

Position:
$$\frac{\partial \mathbf{P}}{\partial s} = \mathbf{R}\mathbf{v} \tag{1}$$

Orientation:
$$\frac{\partial \mathbf{R}}{\partial s} = \mathbf{R}\hat{\mathbf{u}} \tag{2}$$

Force balance:
$$\frac{\partial \mathbf{n}}{\partial s} = -\rho A\mathbf{g} - \mathbf{f} \tag{3}$$

Moment balance:
$$\frac{\partial \mathbf{m}}{\partial s} = -\frac{\partial \mathbf{P}}{\partial s} \times \mathbf{n} - \mathbf{l} \tag{4}$$

Stretch/shear constitutive relation:
$$\mathbf{v} = \mathbf{v}^* + \mathbf{K}_{\mathbf{se}}^{-1}\mathbf{R}^T\mathbf{n} \tag{5}$$

Bend/twist constitutive relation:
$$\mathbf{u} = \mathbf{u}^* + \mathbf{K}_{\mathbf{bt}}^{-1}\mathbf{R}^T\mathbf{m} \tag{6}$$

The unknown solution is the arc-length-dependent Cosserat state $\mathcal{Y}(s)$, which comprises the catheter centerline position, orientation, internal force, and internal moment. Specifically, $\mathbf{P}(s)$ denotes the centerline position in Cartesian coordinates, $\mathbf{R}(s)$ is the rotation matrix describing material orientation, and $\mathbf{n}(s)$ and $\mathbf{m}(s)$ are the internal force and moment vectors, respectively. The local shear-extension and curvature variables, $\mathbf{v}(s)$ and $\mathbf{u}(s)$, are determined from the constitutive relations in Eqs. (5) and (6). The distributed force and torque are denoted by $\mathbf{f}$ and $\mathbf{l}$. The PINN approximates these coupled

TABLE I
PARAMETERS REQUIRED TO SOLVE THE STATIC PINN PROBLEM

Symbol	Unit	Definition	Value
L	m	Catheter length	0.2
A	m^2	Cross-sectional area	10^{-6}
ρ	kg/m^3	Material density	1000
$\mathbf{K_{se}}$	N	Shear stiffness matrix	$\mathrm{diag}(10, 10, 10)$
$\mathbf{K_{bt}}$	Nm^2	Bending stiffness matrix	$10^{-1}\mathrm{diag}(1, 1, 1)$
$\mathbf{v}^*$	-	Value of v for straight rod	$[0, 0, 1]^T$
$\mathbf{u}^*$	1/m	Value of u for straight rod	$[0, 0, 0]^T$
$\mathbf{g}$	m/s^2	Gravitational acceleration	9.81

state variables, while the position error reported in this study is calculated from the predicted centerline $\mathbf{P}(s)$. The physical parameters used to solve the BVP are listed in Table I.

The resultant equations form a boundary value problem (BVP) which, under a specified end wrench, yields the centerline position and orientation fields that satisfy the applied force and moment boundary conditions. These parameters are subsequently used within the PINN framework.

IV. IMPLEMENTATION OF THE SOLVERS

A. PINN Formulation and Network

The PINN framework employs a function-approximation paradigm, leveraging a deep neural network (DNN). The physical laws are embedded via automatic differentiation to compute exact derivatives, which are then enforced through a composite loss function as follows.

Total loss:

$$L_{total} = \lambda_{phys} \cdot L_{phys} + \lambda_{bc} \cdot L_{bc} + \lambda_{quat} \cdot L_{quat} \quad (7)$$

Physics residual loss:

$$\begin{aligned} L_{\text{phys}} = \lambda_p \cdot \frac{1}{N_c} \sum \left(\left\| \frac{d\mathbf{P}}{ds} - \mathbf{R} \cdot \mathbf{v} \right\|^2 \right) \\ + \lambda_R \cdot \frac{1}{N_c} \sum \left(\left\| \frac{d\mathbf{R}}{ds} - \mathbf{R}\hat{\mathbf{u}} \right\|^2 \right) \\ + \lambda_n \cdot \frac{1}{N_c} \sum \left(\left\| \frac{d\mathbf{n}}{ds} + \rho A \mathbf{g} + \mathbf{f} \right\|^2 \right) \\ + \lambda_m \cdot \frac{1}{N_c} \sum \left(\left\| \frac{d\mathbf{m}}{ds} + \frac{d\mathbf{P}}{ds} \times \mathbf{n} + \mathbf{l} \right\|^2 \right) \end{aligned} \quad (8)$$

Boundary condition loss:

$$\begin{aligned} L_{bc} = \ ||p(0) - \ p_0||^2 + \ ||R(0) - \ I||^2 \\ + ||n(L) - \ n_{tip}||^2 + \ ||m(L) - \ m_{tip}||^2 \end{aligned} \quad (9)$$

Quaternion constraint loss (3D only):

$$L_{\text{quat}} = \tfrac{1}{N_c} \sum \left((\|q(s_i)\| - 1)^2 \right) \quad (10)$$

The composite loss penalizes violations of the governing ODE residuals, boundary conditions, and auxiliary constraints, including quaternion unit-norm enforcement in 3D. PDE residuals are averaged over N_c uniformly spaced collocation points, while boundary residuals are evaluated at $s = 0$ and $s = L$. The boundary weight $\lambda_{bc} = 1000$ was selected empirically to balance these terms, with sensitivity examined in Tables V and VII. The loss guides optimization, while accuracy is assessed independently against the BVP reference (Section V). PINN hyperparameters are listed in Table II.

TABLE II
HYPERPARAMETER SET FOR PINN IN 2D AND 3D CASES

Hyperparameter	2D Value	3D Value
Number of hidden layers	4	8
Hidden layer neurons	64	512
Output neurons	10	13
Num. Parameters	13,258	1,846,285
Activation	Tanh	Tanh
$N_{\text{collocation}}$	500	2000
Adam Learning Rate	10^{-3}	5×10^{-4}
Adam Epochs	50,000	25,000
LBFGS History	50	Every 350 iterations
λ_{BC}	1000	1000
λ_{phys}	1.0 (each)	$10.0\lambda_p$, rest 1.0
Optimizer Schedule	Adam, LBFGS	Adam, LBFGS
Output states	$[p, \theta, n, m]$	$[p, q, n, m]$

TABLE III
MESH PARAMETERS FOR THE BVP SOLVER IN 2D AND 3D CASES

Hyperparameter	2D Value	3D Value
BVP Mesh Points	100	200
BVP Max Nodes	5000	5000

Training combines Adam for initial convergence and L-BFGS for high-precision refinement. Collocation points are uniformly distributed along the reference arc length, and their number and loss weights are empirically tuned. The trained PINN supports evaluation at arbitrary domain points, offering a mesh-free, reusable solution for inverse problems and potential real-time applications, at the cost of upfront training and hyperparameter tuning.

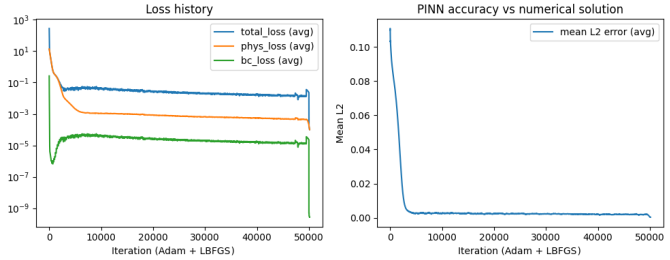
B. BVP Numerical Solver

The reference solution is computed using SciPy's `solve_bvp` with a relative residual tolerance of 10^{-6} and identical equations, parameters, and boundary conditions. It formulates the system as a set of first-order ordinary differential equations derived from force and moment equilibrium and then solves them iteratively over a discrete mesh. The mesh parameters used for the 2D and 3D BVP solutions are reported in Table III. This method requires an initial guess for the state vector and strongly enforced boundary conditions to converge to a high-accuracy solution. While robust and precise, this approach is mesh-dependent and must be recomputed for any change in boundary conditions or physical parameters.

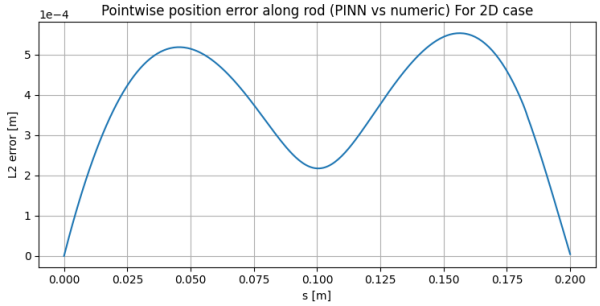
V. RESULTS AND ANALYSIS

The data-free PINN was evaluated in both planar (2D) and spatial (3D) magnetic catheter simulations. The results demonstrate the method's accuracy, robustness, and potential applicability in minimally invasive surgery.

Errors are computed against the BVP reference. With $e_i = \|\mathbf{P}_{\text{PINN}}(s_i) - \mathbf{P}_{\text{ref}}(s_i)\|_2$, the mean error $E_{\text{mean}} = \frac{1}{N}\sum_i e_i$, the tip error $E_{\text{tip}} = e(L)$, and their values normalized by L are reported. The response-time ratio is $t_{\text{BVP}}/t_{\text{PINN}}$, where t_{BVP} is the wall-clock time of one complete `solve_bvp` run and t_{PINN} that of one forward pass of the trained network at the same points. Training is excluded, so the ratio applies to repeated evaluation of an already trained model.



(a) Training convergence for 2D reference case showing total loss and physics residuals (on logarithmic and linear scales, respectively).



(b) Error normalized by catheter length for the 2D reference case.

Fig. 1. Training and error results for the 2D reference case: (a) convergence of total loss and physics residuals; (b) normalized position error along the rod.

A. 2D Case

For the planar configuration, the proposed data-free PINN framework was first validated on the 2D static Cosserat equilibrium problem. The catheter was trained on a reference configuration of length $L = 0.2$ m subjected to a tip force $\vec{F} = (-5\ \mathrm{N})\hat{j}$ and moment $\vec{\tau} = (-2\ \mathrm{N}\cdot\mathrm{m})\hat{k}$, with the training results reported in Fig. 1, where (a) demonstrates the smooth reduction of the total loss and physics residuals over the combined Adam (50,000 epochs) and L-BFGS optimization schedule, and (b) presents the normalized position error along the rod, achieving a sub-millimeter mean position error of 1.13×10^{-4} m ($\approx 0.06\%$ of the rod length) and a tip position error of 3.2×10^{-5} m relative to the BVP reference. Notably, the PINN achieves these results with a solution time approximately 450 times faster than the BVP solver for this reference case.

To elucidate the performance and robustness of the proposed PINN, two summary tables are presented. Table IV details multiple case studies, demonstrating that the network consistently delivers accurate solutions across various configurations, with an average error of 1.48×10^{-4} m and a response time 376 times faster than numerical methods. The corresponding results are visualized in Fig. 2, where all cases are plotted together, clearly demonstrating the agreement between the PINN-predicted and BVP solver reference centerlines and internal loads, even under large curvature conditions induced by external wrenches. Furthermore, Table V reports a hyperparameter study, showing that the selected loss weights and network parameters yield both accurate and stable convergence, whereas inappropriate choices lead to noticeably higher errors.

TABLE IV
PINN PERFORMANCE RELATIVE TO THE BVP REFERENCE ACROSS DIFFERENT CASE STUDIES

Case	E_{mean} (mm)	E_{tip} (mm)	Speed-up ($t_{\mathrm{BVP}}/t_{\mathrm{PINN}}$)	Description
Reference	0.113	0.032	450	L=0.20 m, $\vec{F}$=-5 N$\hat{j}$, $\vec{\tau}$=-2 N·m$\hat{k}$
Case 1	0.231	0.048	126	L=0.20 m, $\vec{F}$=3 N$\hat{j}$, $\vec{\tau}$=1 N·m$\hat{k}$
Case 2	0.165	0.057	995	L=0.30 m, $\vec{F}$=5 N$\hat{j}$, $\vec{\tau}$=0.5 N·m$\hat{k}$
Case 3	0.086	0.009	203	L=0.10 m, $\vec{F}$=-5 N$\hat{j}$, $\vec{\tau}$=-2 N·m$\hat{k}$
Case 4	0.146	0.042	106	L=0.25 m, $\vec{F}$=5 N$\hat{j}$, $\vec{\tau}$=2 N·m$\hat{k}$
Average	0.148	0.038	376	

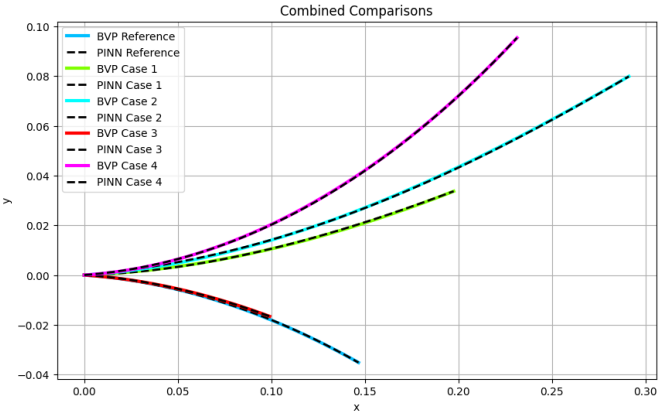
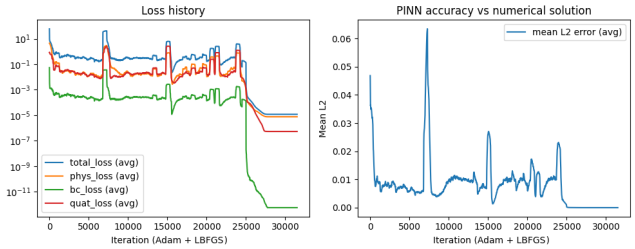


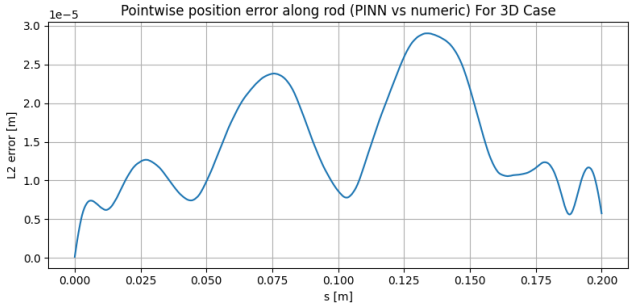
Fig. 2. Comparison of Predicted Centerline Displacements for PINN and BVP Solutions in 2D Cases with Varying Parameters

TABLE V
EFFECT OF HYPERPARAMETER VARIATIONS ON PINN ACCURACY

Hyperparameter Setting	E_{mean} (mm)	E_{tip} (mm)	Notes
Reference	0.113	0.032	Baseline configuration
Reduced Epochs	2.10	0.33	Less learning and Convergence
Changed Loss Weights (λ_{BC}=100, λ_{phys}=10)	4.31	5.58	Large increase in error
Without L-BFGS Fine Tuning	16.029	77.842	Converged to a suboptimal local minimum



(a) Training convergence for 3D reference case showing total loss and physics residuals (on logarithmic and linear scales, respectively).



(b) Error normalized by rod length for the 3D reference case.

Fig. 3. Training and error results for the 3D reference case: (a) convergence of total loss and physics residuals; (b) normalized position error along the rod.

B. 3D Case

Having validated the data-free PINN framework on the 2D static Cosserat equilibrium problem, the analysis is extended to the 3D configuration to assess its performance under more complex spatial dynamics. The framework was applied to a 3D static Cosserat problem with a reference length of $L = 0.20$ m subjected to a tip force $\vec{F} = (-3\ \mathrm{N})\hat{i} + (-3\ \mathrm{N})\hat{j}$ and moment $\vec{\tau} = (1\ \mathrm{N}\cdot\mathrm{m})\hat{i} - (1\ \mathrm{N}\cdot\mathrm{m})\hat{j}$, with the training results reported in Fig. 3, where (a) exhibits the reduction of total loss and physics residuals under a combined Adam (25,000 epochs) and L-BFGS (7000 epochs) optimization schedule, and (b) presents the normalized position error along the rod, achieving a mean position error of 1.48×10^{-5} m ($\approx 0.007\%$ of the rod length) and a tip error of 5.87×10^{-6} m compared to the numerical benchmark. The solution time was about 237 times faster than the BVP solver.

To further assess scalability and robustness, the 3D case results are summarized in Tables VI and VII. Table VI reports performance across multiple 3D studies, confirming that the PINN sustains high accuracy under complex spatial loading, with an average error of 1.82×10^{-5} m and inference speed approximately 246 times faster than the BVP solver. The predicted and reference centerlines, shown in Fig. 4, demonstrate excellent agreement even in cases with strong torsion and bending. Table VII summarizes the hyperparameter analysis for the 3D setup, showing that the selected network depth, learning rate, and loss-weight ratios ensure stable convergence and precise predictions, whereas deviations lead to increased residuals and slower convergence.

TABLE VI
PINN PERFORMANCE RELATIVE TO THE BVP REFERENCE ACROSS DIFFERENT 3D CASE STUDIES

Case	E_{mean} (mm)	E_{tip} (mm)	Speed-up ($t_{\mathrm{BVP}}/t_{\mathrm{PINN}}$)	Description
Reference	0.0148	0.00587	237	L=0.20 m, $\vec{F}$=$-3\ \mathrm{N}\hat{i}-3\ \mathrm{N}\hat{j}$, $\vec{\tau}$=1.0 N·m$\hat{i}-1$ N·m$\hat{j}$
Case 1	0.0535	0.0312	299	L=0.30 m, $\vec{F}$=$-3\ \mathrm{N}\hat{i}-3\ \mathrm{N}\hat{j}$, $\vec{\tau}$=-1 N·m$\hat{j}$
Case 2	0.00783	0.00738	244	L=0.15 m, $\vec{F}$=5 N$\hat{i}$+5 N$\hat{j}$, $\vec{\tau}$=0.5 N·m$\hat{k}$
Case 3	0.00810	0.0110	103	L=0.20 m, $\vec{F}$=3 N$\hat{i}$+3 N$\hat{j}$, $\vec{\tau}$=1 N·m$\hat{j}$
Case 4	0.00688	0.00488	349	L=0.25 m, $\vec{F}$=5 N$\hat{i}$+2 N$\hat{j}$, $\vec{\tau}$=0.5 N·m$\hat{j}$+0.5 N·m$\hat{k}$
Average	0.0182	0.0121	246	

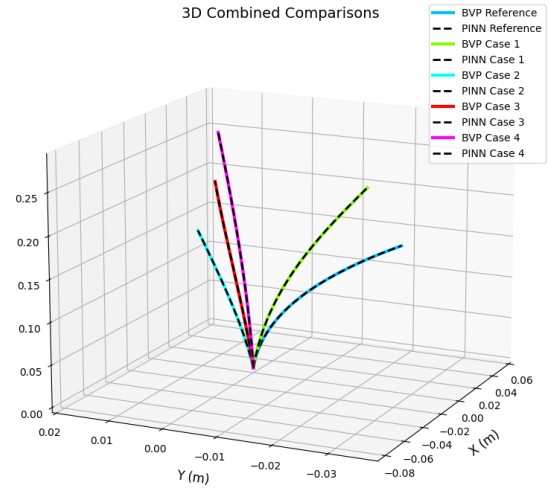


Fig. 4. Comparison of Predicted Centerline Displacements for PINN and BVP Solutions in 3D Cases with Varying Parameters

TABLE VII
EFFECT OF HYPERPARAMETER VARIATIONS ON PINN ACCURACY IN 3D

Hyperparameter Setting	E_{mean} (mm)	E_{tip} (mm)	Notes
Reference	0.0148	0.00587	Baseline configuration
Reduced Layers	0.677	0.883	Reduced learning and learnable parameters
Changed Loss Weights (λ_{BC}=10, λ_{phys}=100, λ_{quat}=1)	20.1	53.32	Increased error due to imbalance
Without L-BFGS Fine Tuning	17.08	52.95	Converged to a suboptimal local minimum

C. Analysis

The case studies in Tables IV and VI illustrate the PINN's robustness across diverse configurations, with E_{mean} values consistently below 0.231 mm in 2D and 0.0535 mm in 3D, and average tip errors of 0.038 mm and 0.0121 mm, respectively. These results span catheter lengths from 0.10 m to 0.30 m and wrench magnitudes up to 5 N and 2 N·m, confirming accurate capture of large deformations without data supervision. Hyperparameter variations, detailed in Tables V and VII, reveal the framework's sensitivity to tuning: baseline settings yield optimal sub-millimeter precision, while reduced epochs or layers increase errors by over an order of magnitude (e.g., 16.029 mm E_{mean} in 2D without L-BFGS). Imbalanced loss weights, such as lowered λ_{BC} or elevated λ_{phys}, lead to convergence issues, emphasizing the need for empirical balancing. A low loss is necessary but not sufficient for accuracy, so E_{mean} was tracked against the BVP reference throughout training and all accuracy claims rest on it. Despite their benefits, PINNs face key limitations. Their rapid inference is offset by a slow, computationally expensive training process that can take hours on modern GPUs, reducing their utility for rapid prototyping. Furthermore, their sensitivity to hyperparameters makes convergence challenging for stiff or highly nonlinear systems. Despite these challenges, meticulous design can leverage PINNs' main strengths.

By leveraging the core advantages of PINNs—being mesh-free and physics-efficient—we constructed a purely physics-driven solver that relies solely on the governing equations and boundary conditions. This performance was achieved through careful empirical tuning, including balancing physics-residual weights, boundary-condition penalties, network architecture design, and optimizer scheduling, to obtain an optimal configuration that converges despite the absence of training data. The framework attained exceptional accuracy, outperforming prior data-dependent approaches in precision and efficiency. However, a direct comparison with [19], [20] is not feasible due to differences in the systems and reference solutions.

VI. Conclusion and Future Work

This study demonstrates a data-free PINN framework that achieves sub-millimeter accuracy in static modeling of magnetically steered catheters, with mean errors of 0.148 mm in 2D and 0.0182 mm in 3D, and inference speeds averaging 376 and 246 times faster than BVP solvers, respectively. By enforcing Cosserat equations without data or meshes, it addresses key MIS challenges, enabling reduced fluoroscopy and enhanced navigation precision.

Future work includes several key directions. A primary next step is to extend the framework to the dynamic Cosserat model to capture transient behaviors; this extension may require incorporation of supervised data, such as sensor measurements, to constrain the time-dependent solution. Experimental validation with physical catheter systems is essential for translating the simulation results toward clinical application. Finally, integrating the trained PINN into real-time control loops, such as model predictive control, could enable closed-loop, low- or no-radiation navigation of magnetic catheters.

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